

SOFTWARE REQUIREMENTS SPECIFICATION

Smart Employee Travel System

Document Version:	1.0	Project Type:	Final Year BE Project
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Technology Stack:	Flutter, Dart, SQLite, Provider, Google Maps SDK, Geolocator API		

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1. INTRODUCTION

1.1 Purpose

The **Smart Employee Travel System** is designed to automate employee transportation management within an organization. It replaces manual trip assignment with a digital platform that manages employees, drivers, vehicles, trip scheduling, payments, and live location tracking.

1.2 Scope

The application provides complete travel management for company employees. Scope includes:

- **User Identity & Access Management:** Multi-role authentication for Admins, Employees, and Drivers.
- **Asset & Entity Management:** Complete CRUD operations for Employees, Drivers, and Vehicles.
- **Trip Logistics:** Automated and manual trip creation, assignment, and status management.
- **Real-Time Navigation & Tracking:** GPS live tracking of drivers during active trips using Google Maps.
- **Security & Verification:** One-Time Password (OTP) verification for boarding confirmation.
- **Financial Oversight:** Invoice generation, payment tracking, and export options (PDF/Excel).
- **Reporting:** Operational and financial reporting dashboards.

1.3 Intended Users

- **Administrator:** Oversees system management, assigns trips, tracks active shuttles, and generates reports.
- **Employee:** View scheduled trips, receive driver details, verify trips via OTP, track live shuttle location, and view travel history.
- **Driver:** View assigned trips, navigate routes via Google Maps, update trip statuses (Start/End), and update live location.
- **Organization Management:** Analyzes monthly travel reports and budget utilization.

1.4 Development Tools & Technology Stack

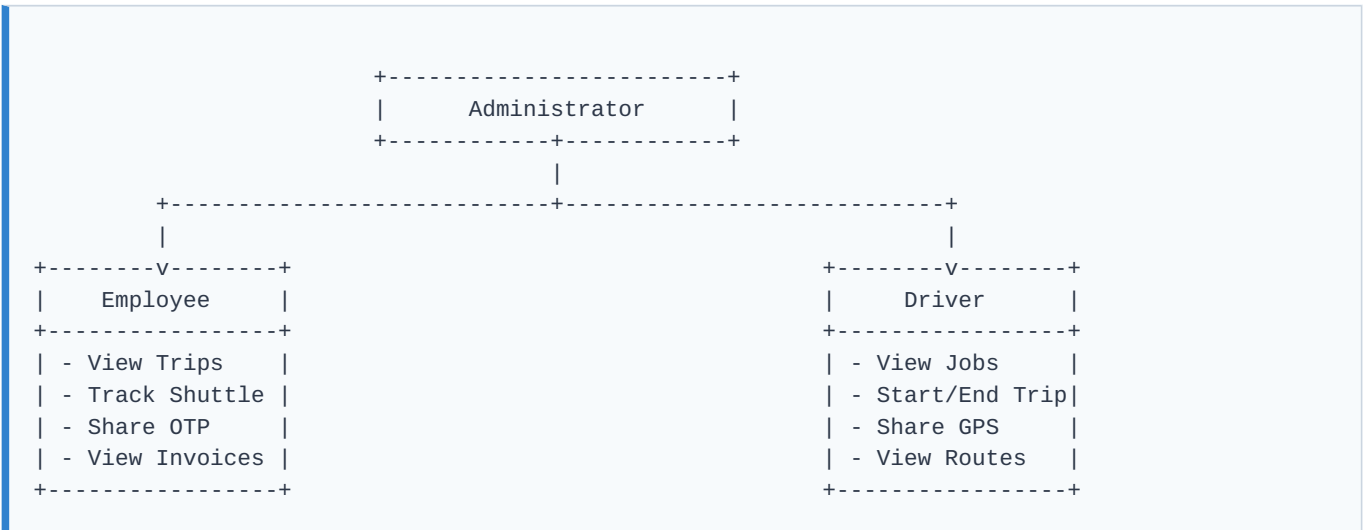
Component	Technology Selected
Frontend UI	Flutter Framework (Dart)
State Management	Provider Package
Local Database	SQLite (sqflite package)
Maps & Location	Google Maps SDK, Geolocator Plugin
IDE & OS	Visual Studio Code / Windows 11

2. OVERALL DESCRIPTION

2.1 Product Perspective

The Smart Employee Travel System operates as a standalone mobile solution utilizing local persistence (SQLite) for offline reliability, combined with real-time location services (Geolocator and Google Maps API).

2.2 User Roles & Interaction



2.3 Operating Environment

- **Mobile OS:** Android 8.0 (API Level 26) or higher / iOS 12.0 or higher.
- **Connectivity:** Cellular Data / Wi-Fi (Required for Google Maps API rendering & live updates).

3. SYSTEM FEATURES & MODULES

3.1 Authentication Module

Provides secure login, registration, forgot password workflow, and OTP verification for safe passenger verification.

3.2 Management Modules

- **Employee Management:** Add, edit, delete, search, and assign home/office locations to employees.
- **Driver Management:** Add, edit, delete, and view driver profiles, license details, and availability status.
- **Vehicle Management:** Registration number, model, capacity limits, vehicle type, and maintenance status tracking.

3.3 Trip & Assignment Module

- **Trip Management:** Create trip, assign driver, assign vehicle, update trip status, view trip history.
- **Assignment Module:** Automatic assignment based on route/capacity and manual override assignment by Admin.

3.4 Live Tracking Module

Integrates GPS tracking to stream driver location during active trips and allow employees to monitor driver progress and route on Google Maps in real-time.

3.5 Payment & Reports Module

- **Payment Module:** Payment dashboard, invoice generation, PDF export, Excel export, share invoice, print invoice.
- **Reports Module:** Daily report, monthly report, payment report, and trip history summary.

4. FUNCTIONAL REQUIREMENTS

ID	Functionality	Description	Priority
FR-01	User Authentication	System shall authenticate users based on username/ password and assigned role.	High
FR-02	Profile Management	Admin shall perform CRUD operations on Employee, Driver, and Vehicle records.	High
FR-03	Trip Dispatch	Admin shall create and assign trips to available drivers and vehicles.	High
FR-04	OTP Verification	System shall generate a 4-digit OTP for the Employee to share with the Driver upon boarding.	High
FR-05	Live Location Streaming	Driver app shall stream GPS coordinates every 5–10 seconds during active trips.	High
FR-06	Map Rendering	Employee app shall plot the driver's live location and destination route on Google Maps.	High
FR-07	Invoice Generation	System shall compute total trip cost and generate downloadable PDF/Excel receipts.	Medium
FR-08	Reporting	Admin shall filter and export daily/monthly travel logs and mileage reports.	Medium

5. NON-FUNCTIONAL REQUIREMENTS

5.1 Performance

App UI screen transitions shall render within 1.5 seconds. GPS updates shall process with latency under 3 seconds under stable network conditions.

5.2 Reliability & Availability

SQLite transactions must maintain ACID properties for local storage. Cached trip information must remain accessible when mobile internet is lost.

5.3 Security

Passcodes and OTPs stored locally must be encrypted. GPS tracking must activate only when a driver starts an official trip and auto-terminate upon trip completion.

5.4 Usability

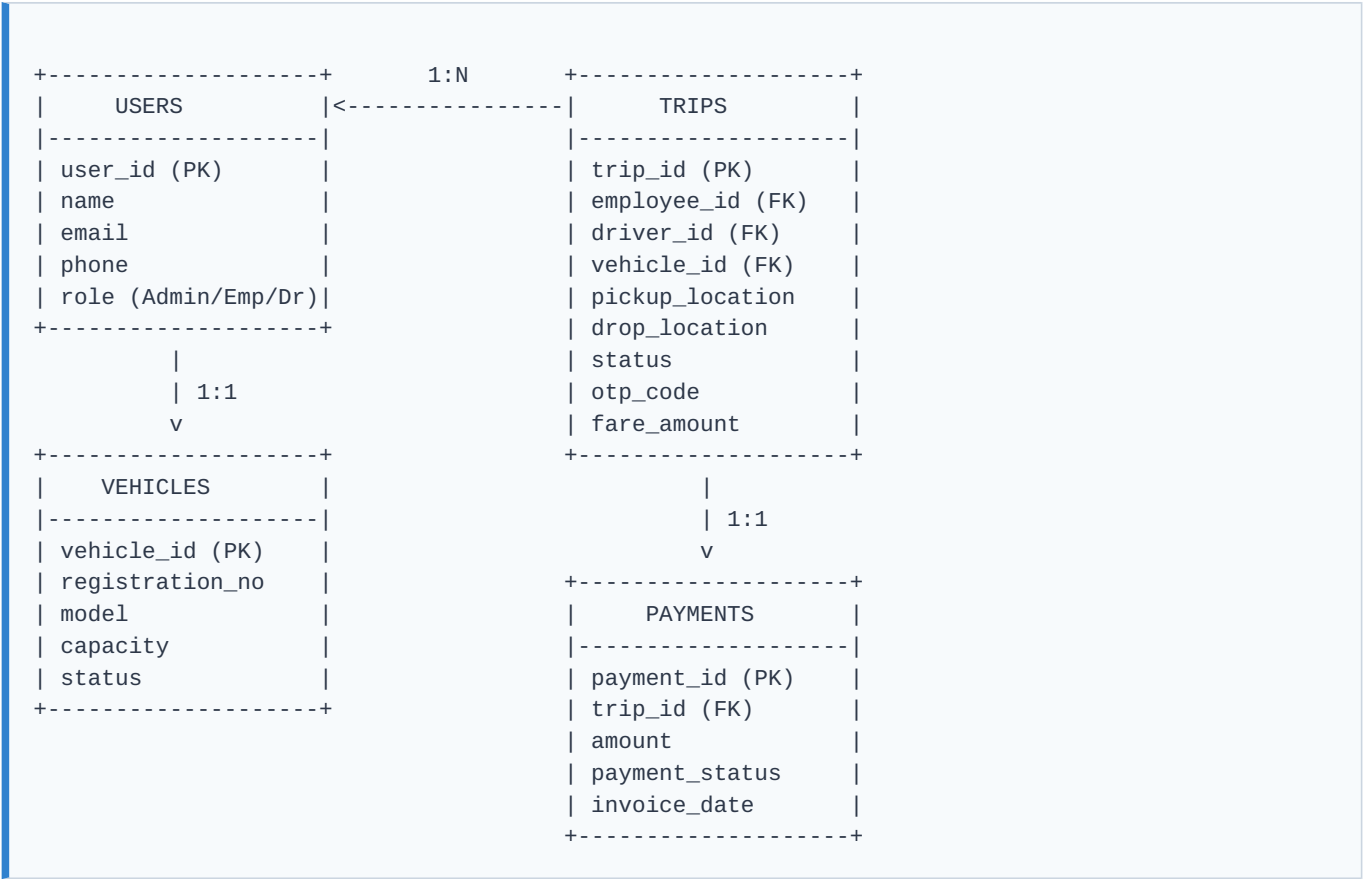
Responsive material design interface optimized for single-handed operation on modern Android/iOS devices.

6. EXTERNAL INTERFACE REQUIREMENTS

- **User Interface (UI):** Clean Flutter Material 3 Design palette with dedicated dashboards customized for Admin, Driver, and Employee views.

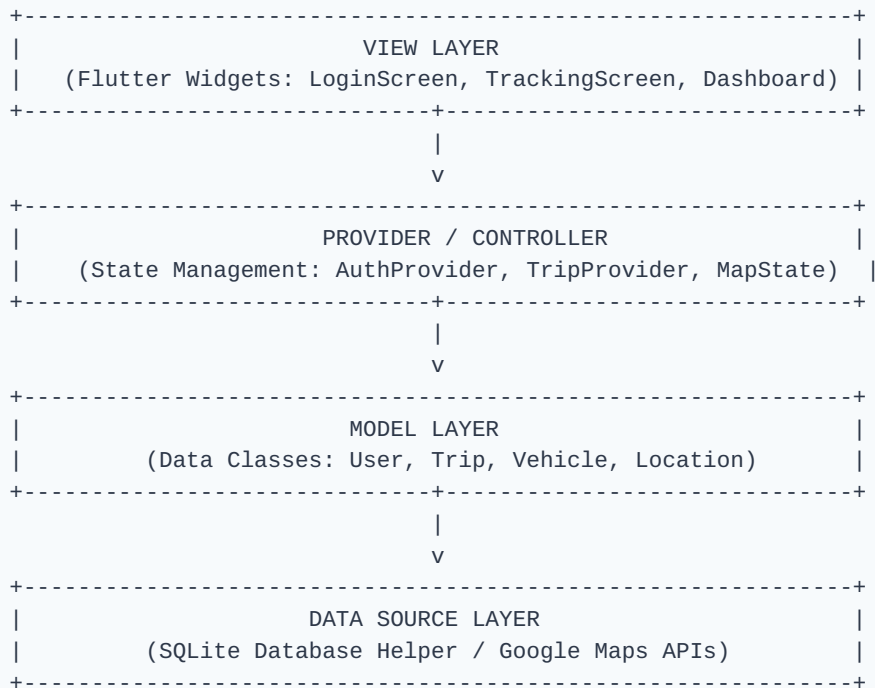
- **Hardware Interfaces:** GPS Receiver accessed via Dart geolocator plugin.
- **Software Interfaces:** Google Maps SDK API for polyline and marker rendering; SQLite C-Library (sqflite) for local data storage.

7. DATABASE DESIGN & SCHEMA



8. SOFTWARE ARCHITECTURE

The application adopts the **MVVM (Model-View-ViewModel)** architectural pattern supported by Flutter’s **Provider** package for state management.



9. HARDWARE & SOFTWARE REQUIREMENTS

9.1 Development Environment Requirements

- **Processor:** Intel Core i5 / AMD Ryzen 5 (8th Gen minimum)
- **RAM:** 8 GB Minimum (16 GB Recommended)
- **Storage:** 256 GB SSD
- **OS:** Windows 11 / macOS / Linux

9.2 Target Mobile Device Requirements

- **Android OS:** Android 8.0 (Oreo) or higher
- **RAM:** 2 GB Minimum
- **Hardware:** GPS / Location hardware sensor enabled
- **Network:** 3G/4G/5G/Wi-Fi connection

10. FUTURE ENHANCEMENTS

1. **AI-Based Route Optimization:** Integrating machine learning algorithms (e.g., Genetic Algorithm or Dijkstra optimization) to automatically construct multi-stop pick-up routes.
2. **Cloud Backend Migration:** Transitioning from local SQLite persistence to Firebase Cloud Firestore / Node.js backend for multi-device network syncing.
3. **Automated Fare Splitting:** Automatic cost-center allocation based on department-wise travel usage.
4. **Emergency / SOS Button:** Instant emergency broadcast system for employee safety during late-night travel.